

MOVIE BOOKING SYSTEM

A Java Swing & JDBC Application

TEAM MEMBERS

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COLLEGE

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GitHub Link:- [Suyash-portfolio\(JDBC-Movie-Booking-System\)](#)

INTRODUCTION



WHAT IS IT?

- ✓ Desktop application built with Java Swing
- ✓ Connects to MySQL via JDBC
- ✓ Allows users to browse movies and book seats
- ✓ Real-time seat availability tracking



PROBLEM STATEMENT

- Manual ticket booking is slow & error-prone
- No real-time seat availability visibility
- Difficult to manage concurrent bookings
- Lack of digital record keeping for users

OBJECTIVES



Easy Ticket Booking

Seamless end-to-end booking from movie selection to seat confirmation



User-Friendly System

Intuitive dark-themed GUI with guided navigation across all screens



Database Management

Structured MySQL schema for users, movies, seats, and bookings



Secure Authentication

Login with prepared statements preventing SQL injection attacks



Real-Time Updates

Live seat availability refreshed from the database on every load



Modular Architecture

Clean separation: Login → Dashboard → Movie → Language → Time → Seat

TECHNOLOGIES USED



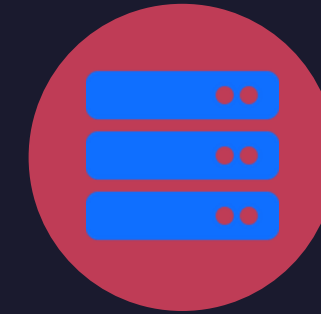
Java Swing

- ✓ Cross-platform GUI framework
- ✓ JFrame, JPanel, JButton
- ✓ Event-driven programming model
- ✓ Custom dark-themed components



JDBC

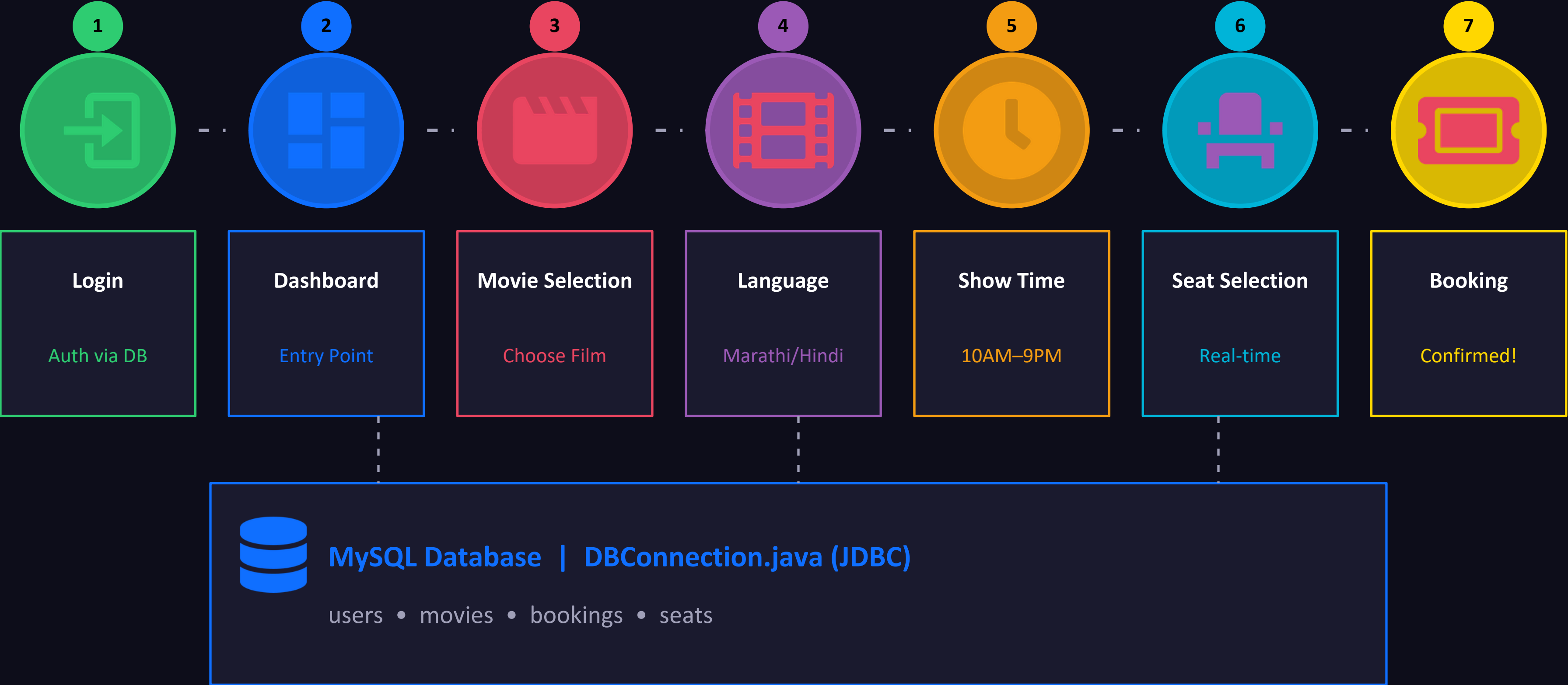
- ✓ Java Database Connectivity
- ✓ PreparedStatement for queries
- ✓ ResultSet for data retrieval
- ✓ Connection pooling ready



MySQL

- ✓ Relational database engine
- ✓ Tables: users, bookings
- ✓ InnoDB for ACID compliance
- ✓ mysql-connector-j driver

SYSTEM ARCHITECTURE



DATABASE DESIGN

 TABLE: users

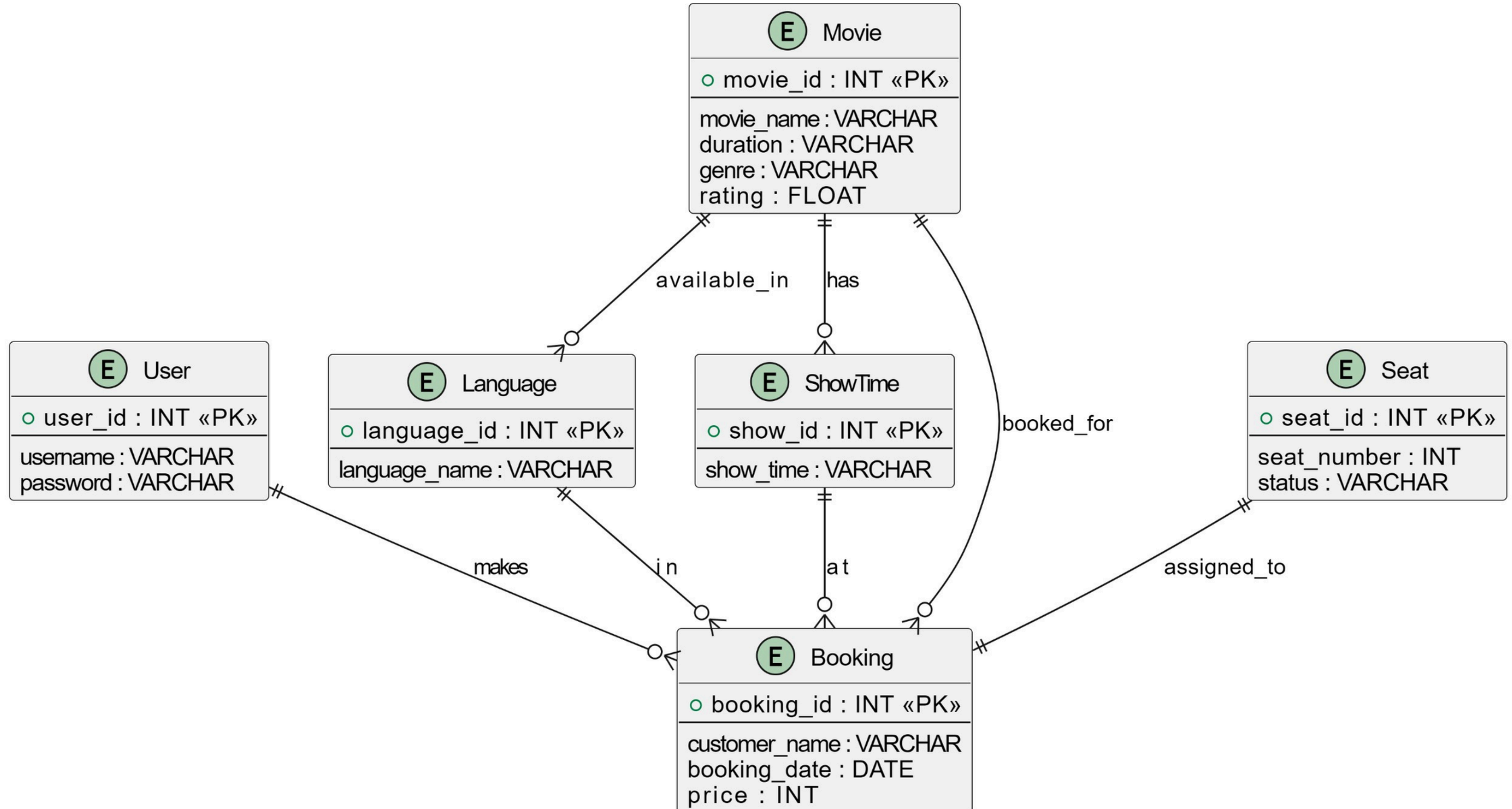
id	INT (PK, AI)	Primary Key
username	VARCHAR(50)	Unique login name
password	VARCHAR(100)	Stored password
email	VARCHAR(100)	User email
created_at	TIMESTAMP	Registration time

1 : N

 TABLE: bookings

id	INT (PK, AI)	Primary Key
user_id	INT (FK)	→ users.id
movie_name	VARCHAR(100)	Movie title
language	VARCHAR(20)	Lang preference
show_time	VARCHAR(20)	Selected time
seat_number	VARCHAR(10)	Booked seat
booked_at	TIMESTAMP	Booking time

Entity Relationship(ER) Diagram



KEY FEATURES



Login Authentication

Secure username/password login using PreparedStatement prevents SQL injection. Sessions managed per JFrame lifecycle.



Movie Selection

Browse available movies with rich card-based UI. Seamless navigation to language and time selection screens.



Seat Booking

Interactive seat map showing available, booked, and selected seats. Visual grid updated from the database live.



Real-Time Availability

Seat status fetched fresh from MySQL on every screen load. Prevents double-booking with DB-level constraints.

CODE EXPLANATION

DBConnection.java

```
String URL =  
"jdbc:mysql://localhost/movie_booking"  
DriverManager.getConnection(  
    URL, DB_USER, DB_PASS  
); // Returns Connection
```

Login.java – PreparedStatement

```
PreparedStatement pst = con  
    .prepareStatement(  
    "SELECT * FROM users"  
    + " WHERE username=?"  
    );
```



PreparedStatement

Parameterized queries prevent SQL injection and improve performance through caching



ResultSet

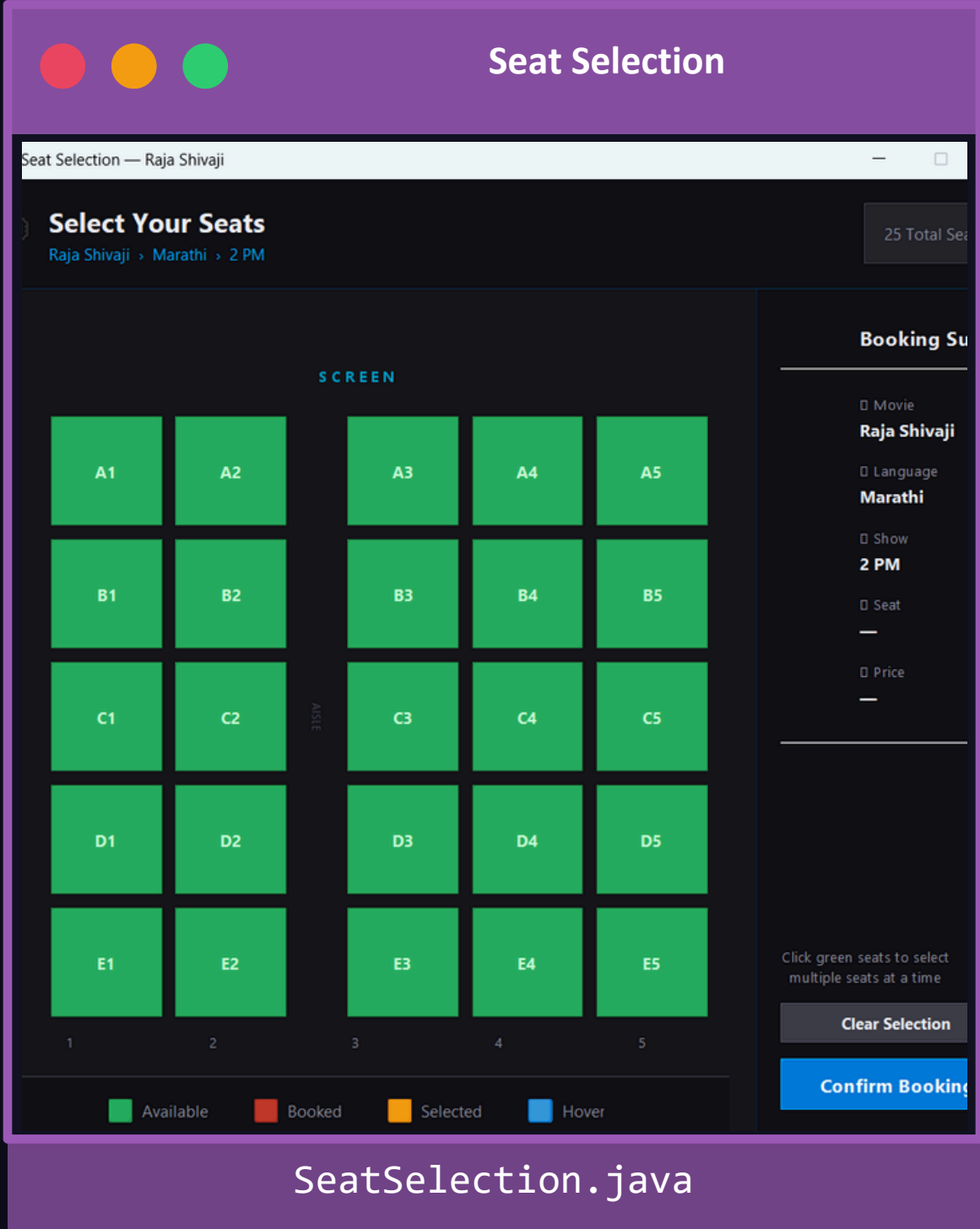
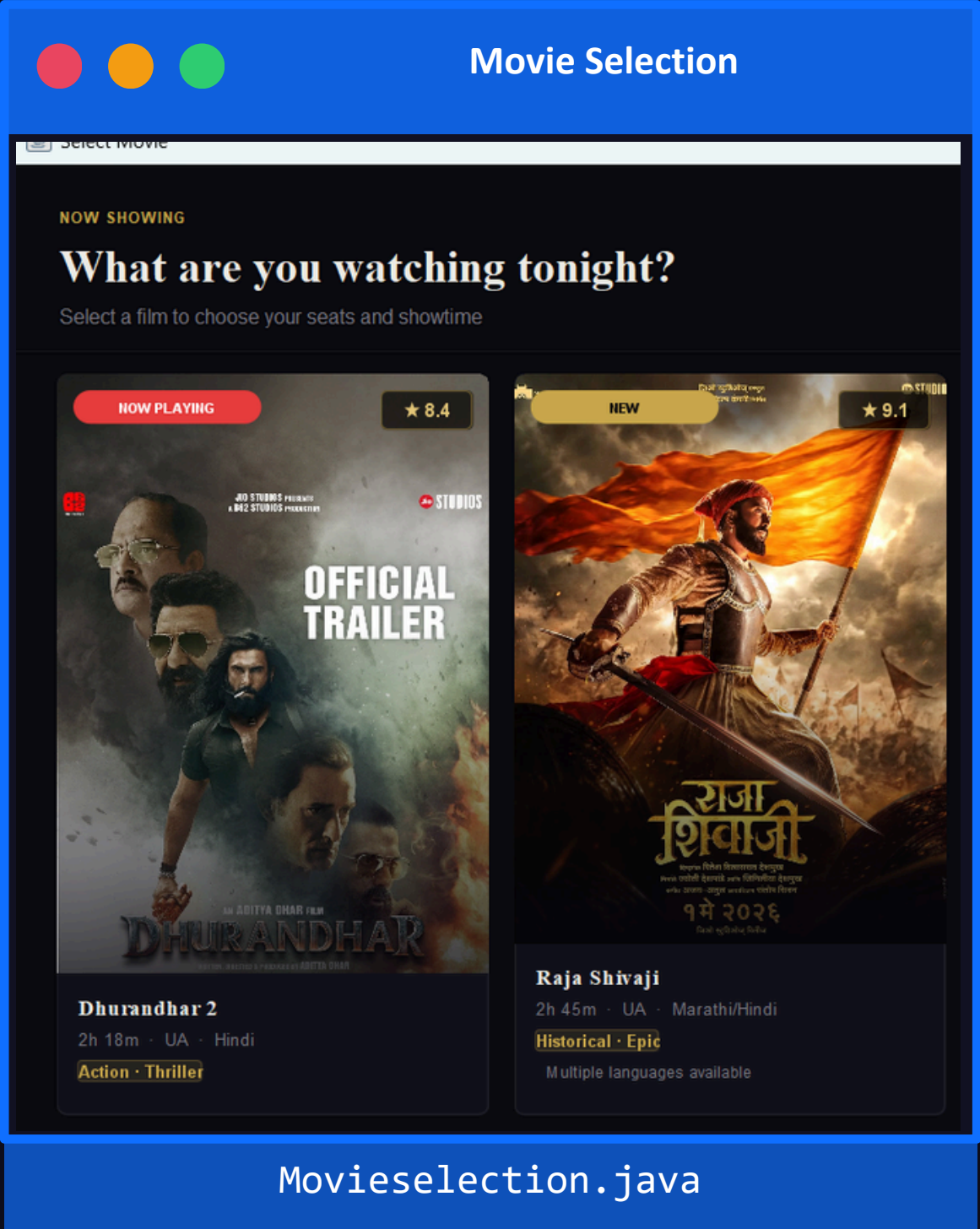
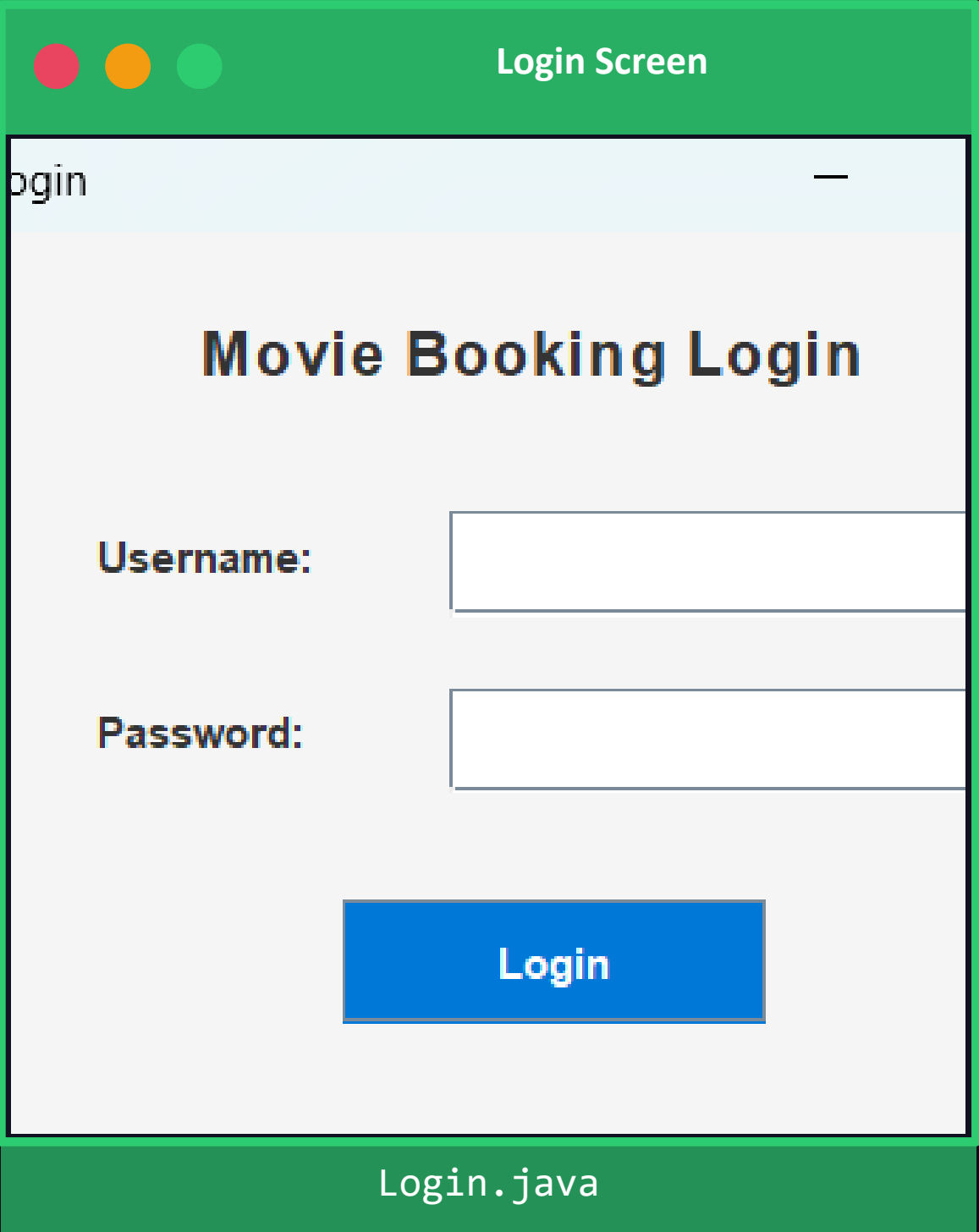
Iterates over query results row by row —
rs.next() moves cursor forward



Connection Lifecycle

DBConnection.isAlive() validates connection before each DB operation

APPLICATION SCREENSHOTS



ADVANTAGES



Easy to Use

5-Step

Booking Flow

Intuitive GUI guides users from login to confirmed booking in under 5 steps



Efficient Booking

< 1s

Response Time

JDBC prepared statements execute fast queries with minimal overhead



Real-Time Updates

Live

Seat Status

Every seat load fetches fresh data — no stale availability issues



Secure System

0

SQL Injection Risk

All queries use PreparedStatement — parameterized inputs are safe



Scalable Design

N:M

Relationship Ready

Relational schema supports adding more movies, users, and showtimes



Clean Architecture

6+

Java Classes

Separated concerns across Login, Dashboard, Selection, and DB layers

CONCLUSION



Project Summary

Successfully developed a Java Swing desktop application with full JDBC-MySQL integration that allows users to authenticate, browse movies, choose language and show time, and book seats with real-time availability tracking.



Future Improvements

- ★ Online payment gateway integration
- ★ Admin panel for movie management
- ★ SMS/email booking confirmation
- ★ Mobile app version (Android/iOS)

THANK YOU

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Java • Swing • JDBC • MySQL

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